

NOBEOKA, Japan

WMO: 478220

Lat: 32.5819N

Lon: 131.6581E

Elev: 21

StdP: 101.07

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.0	0.1	-9.2	1.7	4.2	-7.7	2.0	4.5	10.5	7.9	9.5	7.6	2.2	270	0.346

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.2	32.6	26.0	31.7	25.8	30.8	25.6	27.1	30.6	26.6	30.0	26.2	29.4	3.2	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.6	28.8	25.7	21.0	28.4	25.2	20.4	28.0	85.5	30.7	83.4	30.0	81.5	29.5	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.7	7.2	5.9	-3.3	35.0	1.0	0.9	-4.0	35.6	-4.5	36.1	-5.1	36.6	-5.8	37.3		
			-4.5	27.9	1.1	0.5	-5.4	28.3	-6.0	28.6	-6.6	28.9	-7.5	29.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.2	7.0	8.2	11.2	15.6	19.7	22.7	26.7	27.5	24.6	19.7	14.2	9.0
	DBStd	7.49	2.67	3.24	3.08	2.77	2.02	2.19	1.70	1.34	2.22	2.73	3.27	3.03
	HDD10.0	254	101	69	22	1	0	0	0	0	0	0	4	57
	HDD18.3	1392	352	284	221	88	9	0	0	0	0	17	129	291
	CDD10.0	2882	7	18	60	168	301	381	517	541	437	300	128	25
	CDD18.3	978	0	0	0	6	52	131	258	283	187	58	4	0
	CDH23.3	7678	0	0	2	16	189	647	2381	2852	1340	245	7	0
	CDH26.7	2300	0	0	0	1	13	117	781	1031	336	22	0	0
Wind	WSAvg	2.3	2.7	2.7	2.8	2.4	2.1	1.7	1.9	2.1	2.2	2.3	2.3	2.6
Precipitation	PrecAvg	2376	63	81	158	199	247	387	289	270	331	184	107	61
	PrecMax	3383	157	172	383	357	718	694	861	646	621	579	317	157
	PrecMin	1513	14	24	63	55	100	133	34	40	65	8	17	0
	PrecStd	468	41	48	75	85	132	140	208	149	160	170	74	45
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.6	20.0	22.5	25.2	28.2	32.2	33.9	34.2	32.1	28.7	24.2	19.3
		MCWB	13.1	15.5	15.7	18.9	20.0	25.4	25.8	26.1	25.3	23.3	18.8	14.0
	2%	DB	15.1	17.8	20.2	23.4	26.5	29.5	32.4	32.9	30.7	27.1	22.3	17.5
		MCWB	10.4	13.9	14.1	17.0	19.3	24.2	25.9	26.2	24.7	21.4	17.2	12.9
	5%	DB	13.7	15.7	18.6	22.1	25.2	27.7	31.4	32.0	29.7	25.8	21.1	16.0
		MCWB	9.2	11.4	13.4	16.3	18.9	22.9	25.8	26.1	24.2	20.4	16.6	11.5
	10%	DB	12.3	14.0	17.0	21.0	24.3	26.6	30.5	31.1	28.7	24.7	19.7	14.4
		MCWB	8.1	9.8	12.4	15.8	18.6	22.3	25.6	25.9	23.6	19.6	15.7	10.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.8	17.1	17.8	21.1	23.5	26.2	27.6	27.8	27.0	25.1	20.7	16.5
		MCDB	16.4	18.9	20.8	22.4	25.5	29.6	31.2	31.5	30.0	26.8	22.2	17.5
	2%	WB	12.0	14.9	16.1	19.6	22.2	25.3	26.9	27.2	26.1	23.6	19.1	14.2
		MCDB	13.7	17.1	18.1	21.3	23.9	28.2	30.5	30.8	28.9	25.6	20.7	16.1
	5%	WB	10.1	12.8	14.9	18.3	21.1	24.4	26.4	26.8	25.5	22.2	17.9	12.7
		MCDB	12.7	14.7	17.2	20.6	22.9	26.8	29.8	30.2	28.1	24.3	19.9	14.9
	10%	WB	8.6	10.7	13.4	17.2	20.2	23.6	26.0	26.4	24.9	20.9	16.6	11.2
		MCDB	11.6	13.2	16.2	19.7	22.5	25.6	29.3	29.7	27.3	23.2	18.8	13.8
Mean Daily Temperature Range	5% DB	MDBR	9.5	9.6	9.7	9.6	8.7	6.5	6.6	7.2	7.6	8.8	9.3	9.5
		MCDBR	11.1	11.0	11.2	10.9	10.3	8.5	8.5	8.6	8.5	9.6	9.8	10.6
		MCWBR	7.4	7.3	7.1	6.6	5.8	4.1	3.7	3.7	4.0	5.1	5.9	6.7
	5% WB	MCDBR	9.6	9.5	9.2	8.4	7.5	6.9	7.3	7.4	6.8	6.8	7.6	8.6
		MCWBR	7.5	7.5	7.1	6.2	5.4	3.8	3.6	3.6	3.8	4.4	5.4	6.6
Clear-Sky Solar Irradiance	taub	0.418	0.464	0.529	0.552	0.563	0.570	0.558	0.541	0.516	0.478	0.432	0.406	
	taud	2.113	2.020	1.864	1.861	1.901	1.963	2.034	2.074	2.104	2.123	2.168	2.167	
	Ebn at Noon	771	773	753	757	752	744	751	758	757	753	752	760	
	Edh at Noon	128	155	195	203	197	185	171	162	151	138	120	115	
All-Sky Solar Radiation	RadAvg	2.68	3.22	4.17	4.80	5.00	4.01	4.95	5.00	4.10	3.47	2.74	2.43	
	RadStd	0.22	0.28	0.27	0.51	0.58	0.64	0.62	0.44	0.47	0.39	0.29	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page